

DM545/DM871 – Linear and integer programming

Exercise Sheet 4, Autumn 2026

Exercises with the symbol $^+$ are to be done at home before the class. Exercises with the symbol * will be tackled in class and should be at least read at home. The remaining exercises are left for self training after the exercise class.

Exercise 1 $^+$

Using the bounding method we showed in class that

$$\text{the dual of } \max\{c^T x \mid Ax = b, x \geq 0\} \text{ is } \min\{y^T b \mid y^T A \geq c^T\}.$$

Using the same method show that

$$\text{the dual of } \max\{c^T x \mid Ax \leq b, x \geq 0\} \text{ is } \min\{y^T b \mid y^T A \geq c^T, y \geq 0\}.$$

Exercise 2 *

Consider the following LP problem:

$$\begin{aligned} \max \quad & 2x_1 + 3x_2 \\ & 2x_1 + 3x_2 \leq 30 \\ & x_1 + 2x_2 \geq 10 \\ & x_1 - x_2 \leq 1 \\ & x_2 - x_1 \leq 1 \\ & x_1 \geq 0 \end{aligned}$$

- Write the dual problem
- Using the optimality conditions derived from the theory of duality, and without using the simplex method, find the optimal solution of the dual knowing that the optimal solution of the primal is $(27/5, 32/5)$.

Exercise 3

Consider the problem

$$\begin{aligned} \text{maximize} \quad & 5x_1 + 4x_2 + 3x_3 \\ \text{subject to} \quad & 2x_1 + 3x_2 + x_3 \leq 5 \\ & 4x_1 + x_2 + 2x_3 \leq 11 \\ & 3x_1 + 4x_2 + 2x_3 \leq 8 \\ & x_1, x_2, x_3 \geq 0 \end{aligned}$$

Without applying the simplex method, how can you tell whether the solution $(2, 0, 1)$ is an optimal solution? Is it? [Hint: consider consequences of Complementary slackness theorem.]

Exercise 4

Consider the following LP:

$$\begin{aligned} \min \quad & 3x_1 + 2x_2 - 4x_3 \\ & 2x_1 + x_2 + x_3 \geq 3 \\ & x_1 + x_2 + 2x_3 \leq 5 \\ & x_1, x_2, x_3 \geq 0 \end{aligned}$$

Find the optimal solution knowing that the solution of the dual problem is $(u_1, u_2) = (10/3, 11/3)$.

Exercise 5*

Write the dual of the following problem

$$\begin{aligned}
 (P) \quad & \max \sum_{j \in J} \sum_{i \in I} r_j x_{ij} \\
 & \sum_{j \in J} x_{ij} \leq b_i & \forall i \in I \\
 & \sum_{i \in I} x_{ij} \leq d_j & \forall j \in J \\
 & \sum_{i \in I} p_i x_{ij} = p_j \sum_{i \in I} x_{ij} & \forall j \in J \\
 & x_{ij} \geq 0 & \forall i \in I, j \in J
 \end{aligned}$$

Exercise 6* Level Terrain for a New Highway

The Highway Department is planning a new 10-km highway on uneven terrain as shown by the profile in Figure 1. The width of the construction terrain is approximately 50 meters and the differences in height measured in meters can be deduced from the y -axis in the figure. To simplify the situation, the terrain profile can be replaced by a step function as shown in the figure. Using heavy machinery, earth removed from high terrain is pulled with effort to fill low areas. There are also two burrow pits, I and II, located at the ends of the 10-km stretch from which additional earth can be pulled, if needed. Pit I is located before km 0 and has a capacity of 15 000 cubic meters, and pit II is located after km 10 and has a capacity of 11 000 cubic meters. The costs of removing earth from pits I and II are, respectively, 15 kr and 19 kr per cubic meter.

The transportation cost per cubic meter per kilometer is 1.5 kr, and the cost of using heavy machinery to load pulling trucks is 2 kr per cubic meter. This means that, for example, three cubic meters extracted from pit I and transported for 1 km will cost a total of $15 \cdot 3 + 1.5 \cdot 3 \cdot 1 + 2 \cdot 3 = 96$ kr and three cubic meters pulled 1 km away from a hill to a fill area will cost $1.5 \cdot 3 \cdot 1 + 2 \cdot 3 = 55.5$ kr.

Write a linear programming model for finding the minimum cost plan for leveling the 10-km stretch.

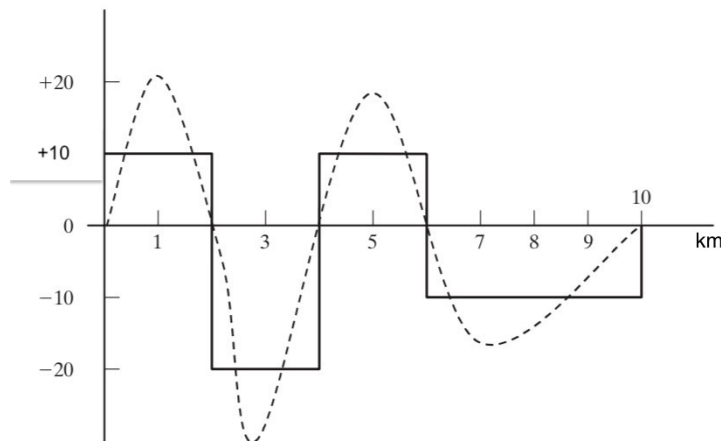


Figure 1:

Exercise 7*

Consider the following problem:

$$\begin{aligned}
 & \text{maximize} && z = x_1 - x_2 \\
 & \text{subject to} && x_1 + x_2 \leq 2 \\
 & && 2x_1 + 2x_2 \geq 2 \\
 & && x_1, x_2 \geq 0
 \end{aligned}$$

In the ordinary simplex method this problem does not have an initial feasible basis. Hence, the method has to be enhanced by a preliminary phase to attain a feasible basis. Traditionally we talk about a *phase I–phase II* simplex method. In phase I an initial feasible solution is sought and in phase II the ordinary simplex is started from the initial feasible solution found.

Phase I can be done by solving an auxiliary LP problem defined by introducing auxiliary variables and minimizing them in the objective. The solution of the auxiliary LP problem gives an initial feasible basis or a proof of infeasibility.

Apply the phase I–phase II simplex method (that is, phase I is carried out with the auxiliary problem) to the problem above.